

5a	When <u>two waves meet at a point</u> , the resultant displacement is equal to the <u>vector sum of the individual displacements</u> .	B1 B1
5b(i)	For observable interference pattern, the <u>two sources must be coherent</u> . At the single slit, the monochromatic light <u>diffracts</u> from a <u>point</u> source. At the double slit, the <u>same diffracted wave</u> reaches S_1 and S_2 (to create two coherent sources).	B1 B1 B1
5b(ii)	Since the path/phase difference is zero and the 2 waves arrive at Y in phase, <u>constructive interference</u> happens at Y.	M1 A1
5b(iii)	From $x = \lambda D / a$ $\lambda = ax / D$ $= (2.0 \times 10^{-3} \times 0.6 \times 10^{-3}) / 1.8$ $= 6.67 \times 10^{-7} \approx 6.7 \times 10^{-7} \text{ m}$	C1 A1
5biv1	the fringe separation will remain the same intensity of bright fringe decreases, while dark fringe remains dark OR less contrast between bright and dark fringes	B1 B1
5biv2	the zeroth order will shift away from Y towards the top of the page Fringe separation / intensity / contrast will remain the same	B1 B1

6a	From Fig. 6.2, magnetic flux density = 50 mT	M1
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